

KRISHNA CHARAN THOTA

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SUMMARY

Software Engineer with 4+ years of experience building distributed, cloud-native backend systems at scale. Proven track record of end-to-end ownership across data ingestion pipelines, API design, webhook delivery systems, and observability infrastructure on **AWS** and **Azure**. Experienced in **Go**, **Java** with hands-on production work spanning **Kubernetes**, **CI/CD** pipelines, and real-time telemetry from **Nokia**, **Starlink** and **SD-WAN** infrastructure. Strong foundation in system reliability, security compliance, and cross-functional collaboration in fast-moving engineering teams.

PROFESSIONAL EXPERIENCE

Software Engineer, Armada AI, Bellevue, WA

Mar 2025 – Present

- Designed and delivered an **end-to-end** cloud-native data ingestion pipeline for **private 5G** and **satellite infrastructure**, integrating **Nokia** and **Starlink** vendor APIs into a unified analytics platform covering radio access nodes, SIM cards, fault alarms, and performance metrics at scale.
- Built a complete **SIM lifecycle management** APIs across multiple service layers with end-to-end ownership from design to production, handling vendor-specific edge cases including partial success states and silent data loss bugs that were causing full SIM visibility loss in production.
- Diagnosed and fixed a production issue where the majority of SIMs were silently excluded from the platform due to a missing access point guard, restoring complete **SIM inventory visibility** without requiring any vendor-side changes.
- Resolved a critical silent data loss bug where edge device alarms were dropped during ingestion due to missing asset associations, implementing a fault-tolerant fallback that eliminated all skipped alarms in production.
- Architected and shipped a distributed **webhook** delivery system with **HMAC** payload signing, **exponential backoff retry**, persistent delivery metadata, async **goroutine-based** dispatch, and automated failure alerting, improving delivery reliability under high-volume alert loads.
- Built a unified observability stack with **Prometheus** and **Grafana** dashboards covering **P99** HTTP latency, per-vendor success rates, in-flight job tracking, and pod-level resource usage across the distributed platform.
- Built a geocoding pipeline integrating **Google Maps API** to convert raw vendor coordinates into human-readable addresses for 5G site and RAN assets, enabling accurate location display across the platform.
- Implemented **audit logging** across the platform covering **webhook lifecycle** events and **SIM provision, deprovision, and edit** actions, providing full traceability for compliance and governance requirements.
- Standardized structured logging and **Prometheus** metrics collection across **distributed services**, establishing consistent observability primitives for production debugging and alerting.
- Implemented automated deployment rollback with health checks and enforced org-wide code review requirements across the **CI/CD** pipeline, improving release safety and engineering accountability.
- Authored full **OpenAPI/Swagger** documentation covering dual versioned API base paths across multiple services, reducing integration friction for cross-functional teams.
- Integrated **Azure Key Vault** with **Managed Identity** for **secret management** and **secure token** retrieval workflows, eliminating hardcoded credentials and strengthening security compliance across services.
- Owned production incident response as primary on-call engineer, driving **high-severity** issue resolution and cross-team escalation coordination to maintain service reliability.

Software Development Engineer, Amazon Services LLC, Bellevue, WA

Oct 2022 – Mar 2025

- Designed a high-performance **Java** API integrated with **DynamoDB**, reducing data retrieval times by **29%** and improving operational efficiency through structured design patterns and best practices.
- Architected **ETL** processes for **1.2 billion** metadata records, consolidating data into a centralized **Redshift** data warehouse, reducing reporting time by **27%** and improving decision-making accuracy.
- Designed a **pre-production** environment with AWS **CodePipeline** and **CDK** in **TypeScript** to validate infrastructure changes before production, minimizing risks through automated monitoring and logging.
- Executed a **throttling** configuration **limiting concurrent** requests during peak traffic, resulting in a **95%** reduction in service disruptions and preventing cascading failures under high load.
- Leveraged AWS services including **API Gateway**, **S3**, **SQS**, **CloudWatch**, **CloudFormation**, **IAM**, **ECS**, and **Fargate** to design, develop, and deploy scalable cloud-native applications.
- Led full **SDLC** across requirement **gathering**, **design**, **testing**, and **deployment** for high-priority **Java** applications, achieving a **98%** on-time delivery rate and consistently meeting delivery timelines across cross-functional teams.
- Drove continuous process improvement as **Scrum Master**, improving team productivity by **15%** through Agile practices and actively contributing to **on-call** rotations.
- Engineered modular Java microservices using **OOP** principles and **Spring Boot**, resulting in a **30%** reduction in debugging time across critical software projects with seamless **NoSQL** database integrations.
- Promoted code quality and maintainability through structured **unit testing** with **JUnit** and **Mockito**, achieving high test coverage across critical service components.

Software Dev Intern, AbhiBus Services India Private Limited, India

Dec 2019 – Apr 2020

- Implemented a robust **Python** application featuring a **HTML** and **CSS** front-end to display extracted **competitor data**, resulting in enhanced comparative analysis capabilities and reducing manual research time by 50 hours each month.
- Engineered a high-performing **machine learning** algorithm in **Python** with a 90.4% accuracy rate for predictive analysis, seamlessly integrating it into the **React.js** for real-time data prediction and display.
- Conducted **factor analysis** and integrated the results into the **React.js** user interface, providing **intuitive data visualization** components for effective **interpretation** and strategic insights.

SKILLS

Programming Languages: Java, Go, Python, TypeScript, JavaScript, Bash, HTML, CSS

Cloud & Databases: AWS (API Gateway, S3, SQS, CloudWatch, CloudFormation, IAM, ECS, Fargate, EC2, CDK), DynamoDB, Redshift, PostgreSQL, MySQL, MongoDB, Oracle, ScyllaDB, Azure (Key Vault, Managed Identity, Workload Identity, AKS)

Frameworks & Libraries: Spring, Spring Boot, Spring Cloud, JUnit, Mockito, React.js, Redux, Jest, Cypress

Observability & Monitoring: Prometheus, Grafana, CloudWatch, Structured Logging, OpenAPI/Swagger

Security & Authentication: JWT (RS256), OAuth2, HMAC, Keycloak, Workload Identity Federation, Azure Key Vault

Containerization: Docker, Kubernetes, Helm, Istio

Development Tools: Git, GitHub, Maven, Jenkins, Jira, Asana, Agile/Scrum, CI/CD, QuickSight

APIs & Architecture: REST, gRPC, Microservices, Distributed Systems

Operating Systems: Linux, Windows

CERTIFICATES

- [AWS Solutions Architect Associate](#)

EDUCATION

Masters of Business Administration, WestCliff University

Sep 2025 – Present

**Master of Professional Studies in Data Sciences and Applications,
University at Buffalo, State University of New York**

Aug 2021 – Aug 2022

**Bachelors, Computer Science and Engineering,
Vignan's Foundation for Science Technology and Research**

May 2016 – May 2020